ಬಿ.ಎಂ.ಎಸ್. ತಾಂತ್ರಿಕ ಮತ್ತು ವ್ಯವಸ್ಥಾಪನಾ ಮಹಾವಿದ್ಯಾಲಯ
BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT
(Autonomous Institution Under VTU)
Yelahanka, Bengaluru -560064

*A Course Project Work Report*

*on*

# "ENTERPRISE SERVICE DESK & INCIDENT MANAGEMENT PLATFORM"

*Submitted in partial fulfillment of the requirements of CCA Phase-2 for*

**BUILDING ENTERPRISE APPLICATIONS (BCS703B)**

in

**INFORMATION SCIENCE AND ENGINEERING**

*By*

**B. VEENA** **USN: 1BY23IS047**

**C. RACHANA** **USN: 1BY23IS055**

*Under the guidance of*

**Mrs. Malini M**
Assistant Professor

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# ABSTRACT

This Phase-2 work turns the Phase-1 design of an Enterprise Service Desk & Incident Management Platform into a working full-stack application for the IT service-management domain. The implemented three-tier architecture uses a React 18 and Bootstrap 5 single-page client served by Nginx, a Spring Boot REST API secured with Spring Security, JSON Web Tokens and BCrypt with three roles (employee, support engineer, administrator), and Spring Data JPA/Hibernate persistence on MySQL 8.4, packaged as a Docker Compose stack. The repository separates frontend, controller, service, repository, model, DTO and security packages, with tests, deployment files, scripts and documentation in their own folders; the deployed backend runs on Spring Boot 4.1.1 with the Chainguard JRE, and the same code is kept as a Spring Boot 3.5.16 reference module. Static analysis with SpotBugs 4.9.8 reported zero bugs, and therefore zero critical bugs; Trivy found no HIGH or CRITICAL vulnerabilities in the deployed backend and frontend container images; Checkstyle (Google Java Style) warnings were reduced from 116 to 0 in both backend modules, and npm audit reports no known vulnerabilities after the frontend build toolchain was upgraded. Testing comprises two JUnit 5 unit tests of the dashboard metrics and six Spring MockMvc API tests covering input validation, authentication, authorization and the incident endpoints; all eight passed in the recorded run. The complete login, create, assign, progress, resolve and report workflow was demonstrated on the deployed Docker stack against MySQL, and a 20-step end-to-end smoke test and a single verification script were added to repeat these checks. The operational outcome is a role-based incident workflow with audit history, in-app notifications and administrator reporting, with the review findings on secrets, demo access, schema safety and report accuracy corrected in the source.

# REPOSITORY SETUP & CODE STRUCTURE

## 1.1 Repository Details

- **Repository Link: https://github.com/<username>/enterprise-service-desk**

  The repository is maintained with Git on the main and dev branches (Figure 1.1).

- **Branching Practice:** main holds stable, verified code; dev is the integration branch; each module or fix is developed on a feature/<name> branch and merged into dev through a pull request; dev is merged into main only after scripts/verify-all passes. .env files, database backups and tokens are never committed.

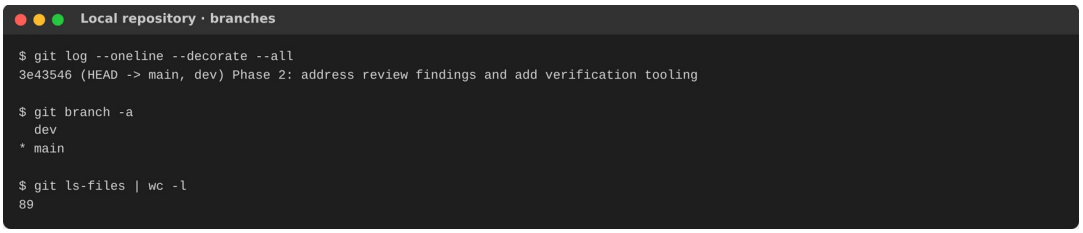



*Figure 1.1: Local repository with main and dev branches*

## 1.2 Technology Environment

| Layer | Technology used |
|---|---|
| Presentation | React 18.3.1, Bootstrap 5.3.8, Vite 7.3.7 (build tool) |
| Web server / proxy | Nginx (alpine): serves the built SPA and forwards /api to the backend |
| REST API | Java 17; Spring Boot 4.1.1 with Tomcat 11.0.26 (deployed); Spring Boot 3.5.16 reference module |
| Security | Spring Security (stateless), JJWT 0.12.6 HMAC-SHA JWT valid 24 h, BCrypt, role and ownership checks |
| Persistence | Spring Data JPA, Hibernate ORM, MySQL 8.4 (H2 in MySQL mode for tests) |
| Testing | JUnit 5, Spring Boot Test + MockMvc, Maven Surefire; Node.js 22 end-to-end smoke test |
| Code quality | SpotBugs 4.9.8, Checkstyle 9.3 (Google Java Style), npm audit, Trivy image scan |
| Deployment | Docker Compose; multi-stage Dockerfiles; Chainguard JRE (backend), Nginx on Alpine with security updates (frontend) |

### 1.3 Project Directory Structure

```
service-desk-phase2/
├── frontend/                    # React UI: src/main.jsx, src/style.css, package.json
├── backend/                     # Spring Boot 3.5 build of the API (reference)
│   ├── src/main/java/com/bea/servicedesk/
│   │   ├── controller/          # REST endpoints (ApiController)
│   │   ├── service/             # Workflow, access rules, metrics
│   │   ├── repo/                # Spring Data JPA repositories
│   │   ├── model/               # JPA entities and enums
│   │   ├── dto/                 # Validated request records
│   │   └── config/              # Security, JWT, optional demo seed
│   ├── src/test/java/...        # JUnit unit + MockMvc API tests
│   ├── pom.xml, Dockerfile
├── backend-spring4-test/        # Same code on Spring Boot 4.1.1 (deployed)
├── scripts/                     # verify-all, smoke-test, backup-db, pin-images
├── docs/                        # Operations, test cases, review response
├── evidence/                    # Archived and fresh analysis/test output
├── compose.yaml                 # MySQL + backend + frontend
├── compose.secure.yaml          # Override for upgrades: schema changes blocked
├── Dockerfile, nginx.conf       # Frontend image and /api reverse proxy
├── .env.example                 # Required variables, no values
└── README.md                    # Setup instructions and run commands
```

### 1.4 Execution & Setup Commands

The application is built and run with Docker Desktop on Windows (PowerShell). Docker downloads Java, Maven, Node.js, Nginx and MySQL inside the containers, so nothing else needs to be installed.

**1. Configure secrets (one time):**

```
cd "F:\VIISem\Enterprise_Service_Desk_Phase2_Final\service-desk-phase2"
Copy-Item .env.example .env
notepad .env   # DB_PASSWORD, MYSQL_ROOT_PASSWORD, JWT_SECRET, JPA_DDL_AUTO=update,
               # DEMO_SEED=true, DEMO_PASSWORD (12+ characters)
```

**2. Build and start MySQL, the Spring Boot API and the React/Nginx frontend:**

```
docker compose up --build -d
docker compose ps
docker compose logs backend --tail 20   # wait for "Started ServiceDeskApplication"
```

**3. Open the application:** http://localhost:8088 (sign in as admin@demo.local, engineer@demo.local or employee@demo.local with DEMO_PASSWORD).

**4. Check the data stored in MySQL:**

```
"SELECT id,title,status FROM incidents;" |
  docker compose exec -T db sh -c 'exec mysql -t -uroot -p$MYSQL_ROOT_PASSWORD servicedesk'
```

**5. Stop the application:**

```
docker compose down        # keeps the data
docker compose down -v     # also deletes the database volume
```

# STATIC CODE ANALYSIS

- **Tools Used:** SpotBugs 4.9.8 (bytecode bug-pattern analysis); Checkstyle 9.3 with google_checks.xml, the version used by maven-checkstyle-plugin 3.5.0; npm audit for frontend dependencies; Trivy for the deployed backend and frontend container images (operating-system packages and libraries).
- **Quality Metrics Summary:**
  - Critical Bugs: **0** (SpotBugs 4.9.8: 0 bug instances of any rank across 7 packages and 32 classes)
  - Vulnerabilities: **0** (Trivy: 0 HIGH/CRITICAL in the deployed backend and frontend images; npm audit: 0 in frontend dependencies after upgrading Vite to 7.3.7)
  - Code Warnings: **0** (Checkstyle 9.3, Google Java Style; reduced from 116 in the original code to 0 in both backend modules)

| Metric | Result | Source |
|---|---|---|
| Bugs (all ranks, incl. critical) | 0 bug instances; 7 packages, 32 classes analysed | SpotBugs 4.9.8 XML report (Spring Boot 4 backend) |
| Dependency vulnerabilities | 0 (previously 1 high, 1 moderate in esbuild via Vite 5.4; fixed by upgrading to Vite 7.3.7) | npm audit (frontend) |
| Style warnings | 116 → 0 in backend/ and backend-spring4-test/ | Checkstyle 9.3, Google Java Style |
| Container image vulnerabilities (HIGH/CRITICAL) | 0 HIGH / 0 CRITICAL in the deployed backend (Spring Boot 4.1, Chainguard JRE) and frontend (Nginx, Alpine 3.24.2) images; previously 14 in the Spring Boot 3.5 backend image and 1 in the frontend image | Trivy image scan (Figure 2.4) |

- **Code Issues Resolved:**

| Detected | File Name | Fix / Refactoring Applied |
|---|---|---|
| AvoidStarImport (21): wildcard imports such as model.* and repo.* | ApiController.java, IncidentService.java, model/*.java | Replaced with explicit imports in the required order |
| NeedBraces (20): if/for bodies without braces | ApiController.java, AccessService.java, IncidentService.java | Added braces; split IncidentService.update() into assign() and changeStatus() without behaviour change |
| MissingJavadocType/Method (52) | All public classes, records and methods | Added Javadoc describing purpose, parameters and access rules |
| Indentation (16): 4-space blocks | config/JwtService.java | Reformatted to Google style; extracted constants |
| npm audit: esbuild ≤ 0.24.2 via Vite 5.4 (GHSA-67mh-4wv8-2f99) | frontend/package.json | Upgraded to Vite 7.3.7, exact versions, package-lock.json, npm ci in the image build |
| Trivy: 9 HIGH, 5 CRITICAL in the Spring Boot 3.5 backend image (Jackson, Tomcat, spring-webmvc, libuuid1) | compose.yaml, backend/pom.xml | Deployed the Spring Boot 4.1 backend on the Chainguard JRE (Tomcat 11.0.26, Jackson 2.22.3/3.2.3): 0 findings; pinned Jackson 2.21.7 and Tomcat 10.1.58 in the reference module |
| Trivy: libexpat 2.8.5 HIGH (CVE-2026-77214) in the frontend image | Dockerfile | Added apk upgrade to the Nginx stage (libexpat 2.9.0): 0 findings |
| Manual review: Spring Boot 4-only imports in Spring Boot 3 tests (compile failure) | backend/src/test/.../ ApiIntegrationTest.java | Used the Spring Boot 3 Jackson and MockMvc packages |
| Manual review: 400/401 turned into 403 when forwarded to /error | config/SecurityConfig.java | Permitted /error and enabled RFC 9457 problem details |

## Verification Proof:

**Checkstyle 9.3 · Google Java Style · 18 Oct 2026**

```
# BEFORE: original uploaded source (backend-spring4-test)
$ java -jar checkstyle-9.3-all.jar -c /google_checks.xml src/main/java
Starting audit...
[WARN] model/Priority.java:3:1: Missing a Javadoc comment. [MissingJavadocType]
[WARN] model/Role.java:3:1: Missing a Javadoc comment. [MissingJavadocType]
[WARN] model/Notification.java:3:27: Using the '.*' form of import should be avoided - jakarta.persistence.*. [AvoidStarImport]
[WARN] model/Notification.java:6:1: Missing a Javadoc comment. [MissingJavadocType]
[WARN] model/Incident.java:3:27: Using the '.*' form of import should be avoided - jakarta.persistence.*. [AvoidStarImport]
[WARN] model/Incident.java:6:1: Missing a Javadoc comment. [MissingJavadocType]
... 110 more warnings ...
Audit done.
Warnings: 116

# AFTER: fixed source (backend), run 2026-10-18T13:15:52+00:00
$ java -jar checkstyle-9.3-all.jar -c /google_checks.xml backend/src/main/java
Starting audit...
Audit done.
Warnings: 0

# AFTER: fixed source (backend-spring4-test), run 2026-10-18T13:15:59+00:00
$ java -jar checkstyle-9.3-all.jar -c /google_checks.xml backend-spring4-test/src/main/java
Starting audit...
Audit done.
Warnings: 0
```

*Figure 2.1: Checkstyle 9.3 — 116 warnings on the original source, 0 after refactoring in both backend modules*

**SpotBugs summary (archived XML)**

```
# Archived report from the 16 Oct 2026 run (before the fixes): evidence/archived-2026-10-16-spring4-run/spotbugsXml.xml
SpotBugs version      : 4.9.8
Analysed              : /app/target/classes (backend-spring4-test)
Packages / classes    : 7 / 32
Java                  : 17.0.15
Timestamp             : Fri, 16 Oct 2026 17:15:28 +0000
BugInstance elements  : 0
total_bugs            : 0
```

*Figure 2.2: SpotBugs summary — 0 bugs, 0 critical*

**Frontend: npm ci · vite build · npm audit · 18 Oct 2026**

```
# Run: 2026-10-18T13:16:09+00:00  node v22.22.0  npm 10.9.4
$ npm ci

added 70 packages, and audited 71 packages in 3s

found 0 vulnerabilities
$ npm run build

> enterprise-service-desk@1.0.0 build
> vite build

vite v7.3.7 building client environment for production...
transforming...
✓ 27 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                  0.37 kB │ gzip:  0.27 kB
dist/assets/index-DMN28Om0.css  236.25 kB │ gzip: 32.47 kB
dist/assets/index-nJQ8AcXa.js   158.24 kB │ gzip: 50.26 kB
✓ built in 1.31s
$ npm audit
found 0 vulnerabilities
```

*Figure 2.3: Frontend clean install, production build and npm audit — 0 vulnerabilities*

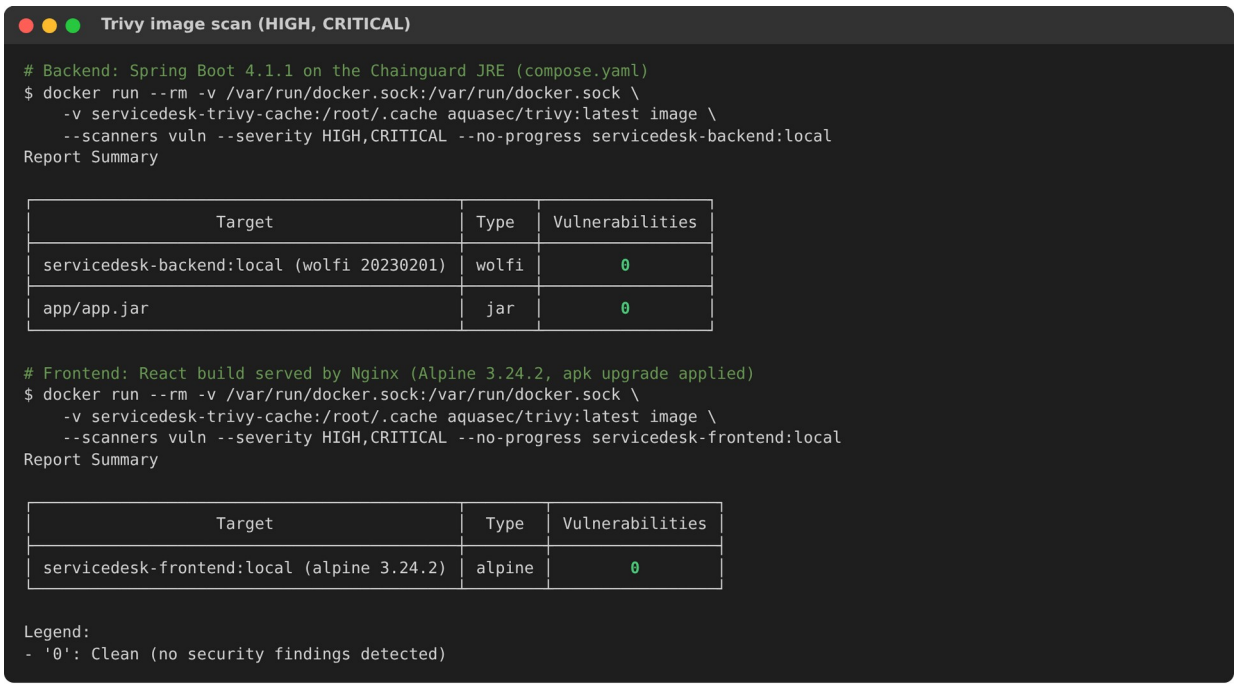



*Figure 2.4: Trivy scans of the deployed images — backend and frontend: 0 HIGH, 0 CRITICAL*

# TESTING EXECUTION

- **Test Framework Used:** JUnit 5 (Jupiter) with Spring Boot Test and Spring MockMvc, run by Maven Surefire against an in-memory H2 database in MySQL mode. The end-to-end workflow was verified on the deployed Docker stack with MySQL (Section 3.1).
- **Test Cases Table:** the eight automated test methods in the source, with the assertion each one makes.

| Test ID | Test Level (Unit / API) | Function / Endpoint Tested | Input / Action | Expected Result | Status (Pass/ Fail) |
|---|---|---|---|---|---|
| TC-01 | Unit | IncidentMetrics.awaitingAction() | OPEN, ASSIGNED and RESOLVED incidents | Returns 2 (only OPEN/ASSIGNED counted) | **Pass*** |
| TC-02 | Unit | IncidentMetrics.completed() | OPEN, RESOLVED and CLOSED incidents | Returns 2 (RESOLVED/CLOSED counted) | **Pass*** |
| TC-03 | API | POST /api/auth/register | Email "invalid" | Status 400 Bad Request | **Pass*** |
| TC-04 | API | POST /api/incidents | Valid token and incident body | Status 201, status = OPEN | **Pass*** |
| TC-05 | API | POST /api/incidents | Blank title | Status 400 Bad Request | **Pass*** |
| TC-06 | API | GET /api/incidents | No token | Status 403 Forbidden | **Pass*** |
| TC-07 | API | GET /api/reports/summary | End-user token | Status 403 Forbidden | **Pass*** |
| TC-08 | API | GET /api/incidents | Valid end-user token | Status 200, JSON array | **Pass*** |

***** *Status from the Maven Surefire reports of the Spring Boot 4 module (8 run, 0 failures, 0 errors; Figure 3.1). The later code clean-up did not change any test assertion. Each API test first registers a new user (POST /api/auth/register, expects 201). Test method names are listed in docs/TEST_CASES_VERIFIED_FROM_SOURCE.md.*

### 3.1 End-to-End Workflow Verification

Login, assignment, status changes, notifications and role restrictions were verified on the deployed Docker stack (Nginx, Spring Boot, MySQL 8.4) using the demo administrator, engineer and employee accounts. The employee signed in and raised a HIGH-priority network incident (Figure 4.3); the administrator assigned it to the support engineer; the engineer moved it to IN_PROGRESS and resolved another ticket with resolution text; and the administrator dashboard shows the live incident counts (Figure 4.2). The incidents and audit events written by these actions were read back from MySQL (Figure 4.4). The same workflow is automated as a 20-check smoke test in scripts/smoke-test.mjs, covering valid and invalid login, validation errors, assignment, progress, resolution rules, audit history, notifications, dashboard counts and access denials.

## Execution Proof:

JUnit 5 / Surefire results (archived XML)

```
# Archived Surefire XML from the 16 Oct 2026 run (Spring Boot 4.1.1, Java 17.0.15, WSL2)

com.bea.servicedesk.IncidentMetricsTest  tests=2 failures=0 errors=0 skipped=0 time=0.132s
  PASS  completedCountsResolvedAndClosed             0.114s
  PASS  awaitingActionCountsOnlyOpenAndAssigned      0.004s

com.bea.servicedesk.ApiIntegrationTest  tests=6 failures=0 errors=0 skipped=0 time=86.565s
  PASS  unauthenticatedListRejected                 14.293s
  PASS  incidentRequiresTitle                        1.221s
  PASS  incidentCreated                              1.064s
  PASS  invalidEmailRejected                         0.220s
  PASS  userCannotAccessAdminReport                  0.303s
  PASS  userCanListOwnIncidents                      0.499s

Total: 8 tests, 0 failures, 0 errors, 0 skipped
```

*Figure 3.1: JUnit 5 results from the Surefire reports — 8 tests, 0 failures, 0 errors*

# CORE IMPLEMENTATION

**Summary of the working features developed:**

- **Module 1 (User Interface / Presentation):** a React single-page application with role-aware navigation for employees, support engineers and administrators. Screens: sign-in and self-registration; an overview dashboard with total, awaiting-action, in-progress and resolved counts and the latest incidents; a searchable incident list; a new-incident form (title, description, category, priority); a ticket detail panel with the activity timeline and an update form (assign engineer, status, resolution, comment); notifications with an unread badge; user-role administration; and status and priority reporting. API errors are shown using the "detail" message returned by the backend.

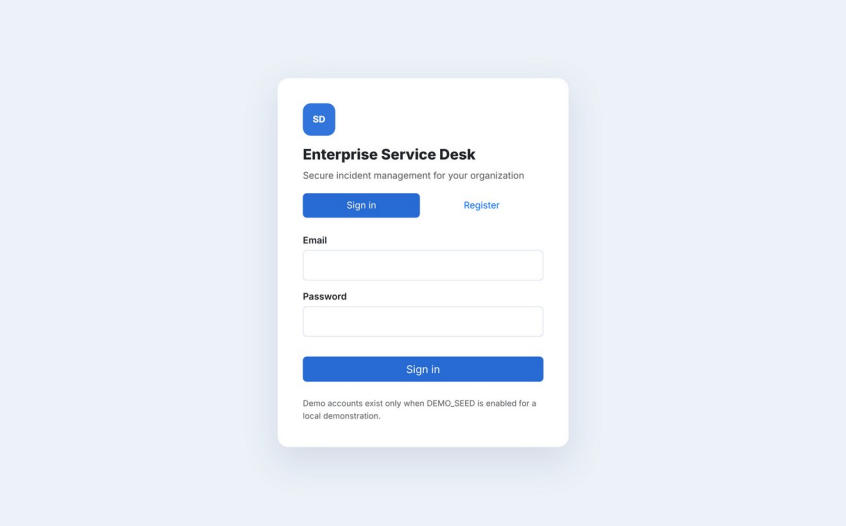



*Figure 4.1: Sign-in screen*

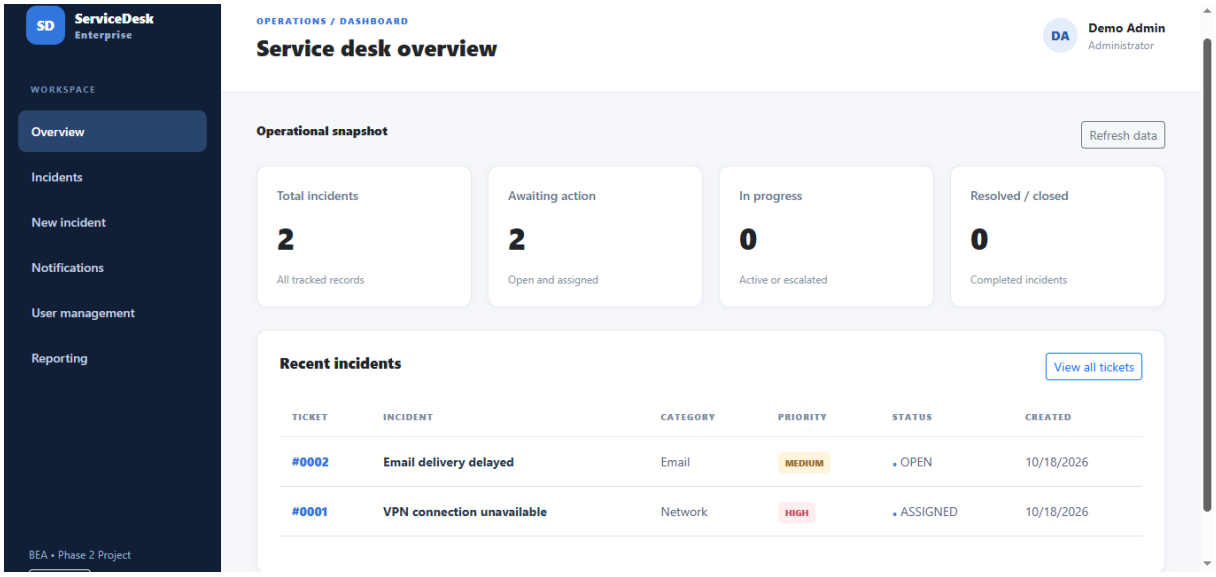



*Figure 4.2: Overview dashboard (administrator)*

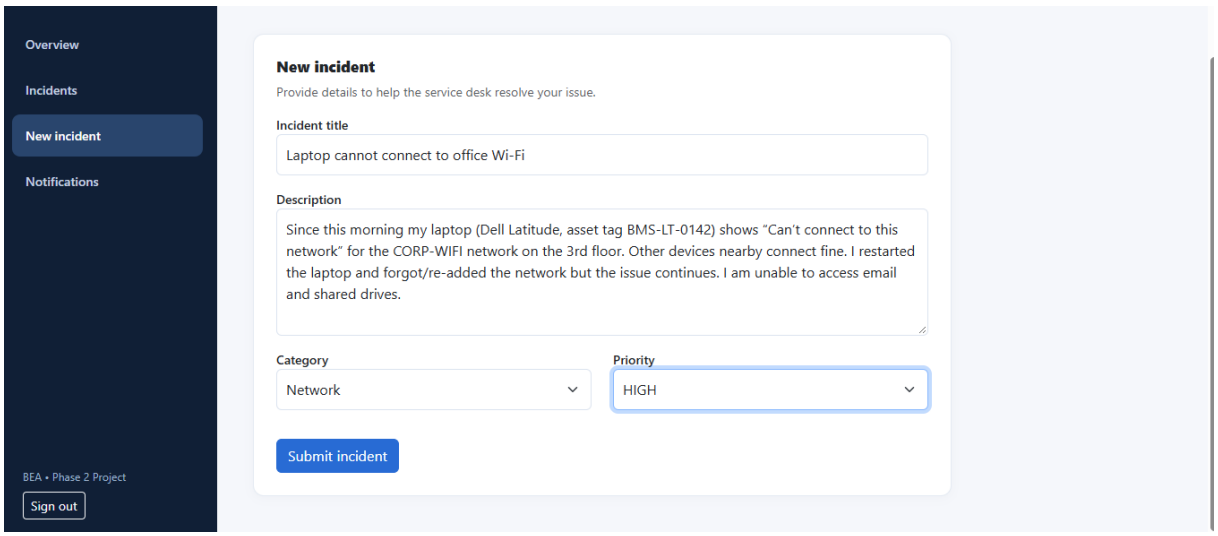



*Figure 4.3: New incident creation form (employee)*

- **Module 2 (Core Business Logic / Processing):** incident workflow management in IncidentService and AccessService. An employee submits an incident; the request is validated and saved as OPEN with a CREATED audit event. An administrator assigns it to a user with the ENGINEER role (status ASSIGNED, assignment event, notification to the engineer). The assigned engineer moves it to IN_PROGRESS or ESCALATED, adds comments and records a resolution, then RESOLVED or CLOSED; each change notifies the requester. Rules enforced on the server:
  - Employees see only their own incidents, engineers only those assigned to them, administrators all.
  - Only employees and administrators create incidents; only engineers and administrators update them; only administrators assign.
  - A ticket cannot be set back to OPEN or ASSIGNED, and RESOLVED/CLOSED require resolution text.
  - Administrators manage roles but cannot change their own role; reports are administrator-only.
  - Each change and its audit events and notifications are saved in one transaction (@Transactional).
- **Module 3 (Data Persistence & Integration):** Spring Data JPA repositories map four entities to MySQL 8.4 tables and use derived queries for role-scoped lists (for example findByAssigneeIdOrderByCreatedAtDesc). Integration is REST/JSON: the browser calls /api on Nginx, which forwards to Spring Boot; every request carries a Bearer JWT that the security filter verifies. No external email or SMS API is connected.

| Table | Key columns | Purpose |
|---|---|---|
| users | id, name, email (unique), password_hash, role | Accounts with BCrypt hashes and one role |
| incidents | id, title, description, category, priority, status, requester_id, assignee_id, resolution, created_at, updated_at | Tickets |
| incident_events | id, incident_id, actor_id, action, detail, created_at | Audit timeline |
| notifications | id, recipient_id, message, read_flag, created_at | In-app notifications |

Schema safety: Hibernate runs with JPA_DDL_AUTO=validate by default, so the application never alters

an existing schema; update is used only once to create a new local database, and the Spring Boot 4 override forces validate. scripts/backup-db takes a mysqldump before any upgrade. Secrets come only from environment variables, the backend refuses a JWT secret shorter than 32 bytes, and demo accounts exist only when DEMO_SEED=true with a DEMO_PASSWORD supplied at run time.

```
PS F:\VIISem\Enterprise_Service_Desk_Phase2_Final\service-desk-phase2>    "SELECT id,title,category,priority,status,
11,40) AS detail,actor_id,created_at FROM incident_events ORDER BY id;" | docker compose exec -T db sh -c 'exec mysq
mysql: [Warning] Using a password on the command line interface can be insecure.
+----+---------------------------------------+----------+----------+-------------+--------------+-------------+
| id | title                                 | category | priority | status      | requester_id | assignee_id |
+----+---------------------------------------+----------+----------+-------------+--------------+-------------+
|  1 | VPN connection unavailable            | Network  | HIGH     | RESOLVED    |            3 |           2 |
|  2 | Email delivery delayed                | Email    | MEDIUM   | OPEN        |            3 |        NULL |
|  3 | Laptop cannot connect to office Wi-Fi | Network  | HIGH     | IN_PROGRESS |            3 |           2 |
+----+---------------------------------------+----------+----------+-------------+--------------+-------------+
+-------------+----------------+-----------------------------------------+----------+----------------------------+
| incident_id | action         | detail                                  | actor_id | created_at                 |
+-------------+----------------+-----------------------------------------+----------+----------------------------+
|           3 | CREATED        | Incident submitted                      |        3 | 2026-10-18 14:34:49.619712 |
|           1 | STATUS_CHANGED | IN_PROGRESS                             |        2 | 2026-10-18 14:41:09.083041 |
|           1 | COMMENT        | Checking the wireless profile and certif |        2 | 2026-10-18 14:41:09.251034 |
|           3 | ASSIGNED       | Demo Engineer                           |        1 | 2026-10-18 14:45:02.089010 |
|           3 | COMMENT        | Laptop cannot connect to office Wi-Fi   |        1 | 2026-10-18 14:45:02.110972 |
|           3 | STATUS_CHANGED | IN_PROGRESS                             |        2 | 2026-10-18 14:47:07.383485 |
|           1 | STATUS_CHANGED | RESOLVED                                |        2 | 2026-10-18 14:48:25.977178 |
|           1 | RESOLUTION     | Reinstalled the VPN client and renewed t |        2 | 2026-10-18 14:48:26.000772 |
|           1 | COMMENT        | VPN access confirmed working with the us |        2 | 2026-10-18 14:48:26.015085 |
+-------------+----------------+-----------------------------------------+----------+----------------------------+
```

*Figure 4.4: Incident and audit-event rows stored in MySQL by the workflow (times in UTC)*

### 4.1 Implementation Boundaries

Not implemented from the Phase-1 target architecture: email/SMS delivery, file attachments, category management, automatic SLA escalation, database replication, monitoring and cloud deployment. Versioned database migrations (Flyway) are the next planned step for production use.

# EVALUATION RUBRICS

**Construction Readiness, Testing and Implementation**

**(60 MARKS Scaled Down to 10 marks)**

| S.No | Evaluation Criteria | Excellent | Good | Needs Improvement | Max Marks | Marks Obtained |
|---|---|---|---|---|---|---|
| 1. | Repo Setup & Code Structure | Well-organized GitHub/GitLab repo setup with modular folder organization. **(9-10 marks)** | Fairly organized repository with minor structural flaws. **(6-8 marks)** | Disorganized repo; lacks clear modular structure. **(0-5 marks)** | 10 | |
| 2. | Static Code Analysis | Tool executed; zero critical bugs; report submitted. **(13-15 marks)** | Tool executed; minor code bugs present. **(9-12 marks)** | Static analysis not performed or results ignored. **(0-8 marks)** | 15 | |
| 3. | Testing Execution | Working Unit and API/Integration tests with passing logs **(17-20 marks)** | Working unit tests, but limited integration test coverage. **(11-16 marks)** | Missing unit/integration tests or execution errors. **(0-10 marks)** | 20 | |
| 4. | Implementation | Functional implementation of core modules matching architectural design. **(13-15 marks)** | Partial implementation with minor functional bugs or gaps **(9-12 marks)** | Incomplete or non-functional module implementation. **(0-8 marks)** | 15 | |
| | | | | **TOTAL MARKS** | | |

Signature of Course Co-Ordinator

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